A Highly PVT Invariant Low Power Scalable Analog Dynamic Current-Mode Translinear Matrix-Vector Multiplier

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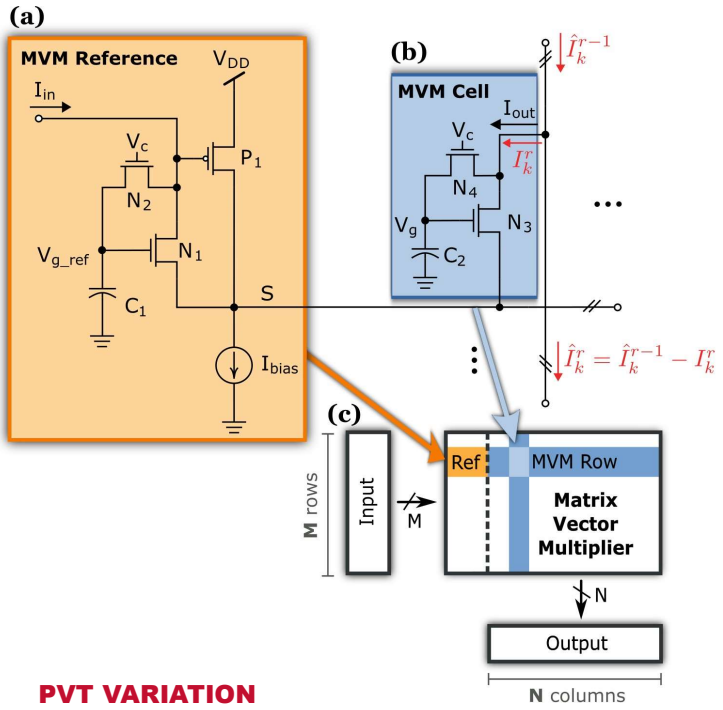
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WHAT'S NEW?

- Enabling precise scalability by **improving robustness** to Process, Voltage and Temperature (PVT) variations.
- Maintaining a high energy efficiency (**0.5fJ per MAC**)

HOW?

- **Dynamic current mirror based design** (offering inherent robustness to PVT)
- **A custom correlated double sampling (CDS) programming scheme** for calibrating and storing the multiplication weights
- Built-in capacitor for on-site analog memory
- Designing in the subthreshold domain



CDS PROGRAMMING SCHEME

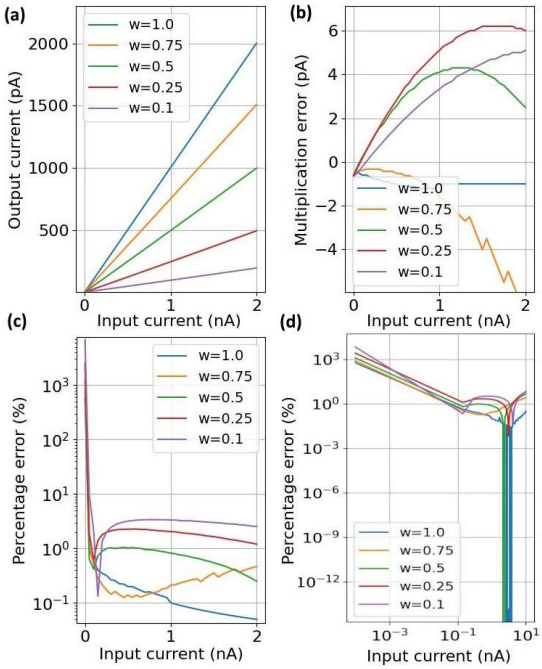
1. The N_2 and N_4 switches turn on
2. We externally set I_{in} and I_{out} so that $I_{out} = w * I_{in}$, where w is the desired weight
3. Wait ample time for both capacitors (each 5 fF) to charge
4. The N_2 and N_4 switches turn off

Now, for a given input, the output maintains the previously set ratio $I_{out} = w * I_{in}$

$$I_{out} = I_{in} \cdot e^{-\kappa \frac{V_g - V_{g\,ref}}{V_T}} = w \cdot I_{in} \quad [1]$$

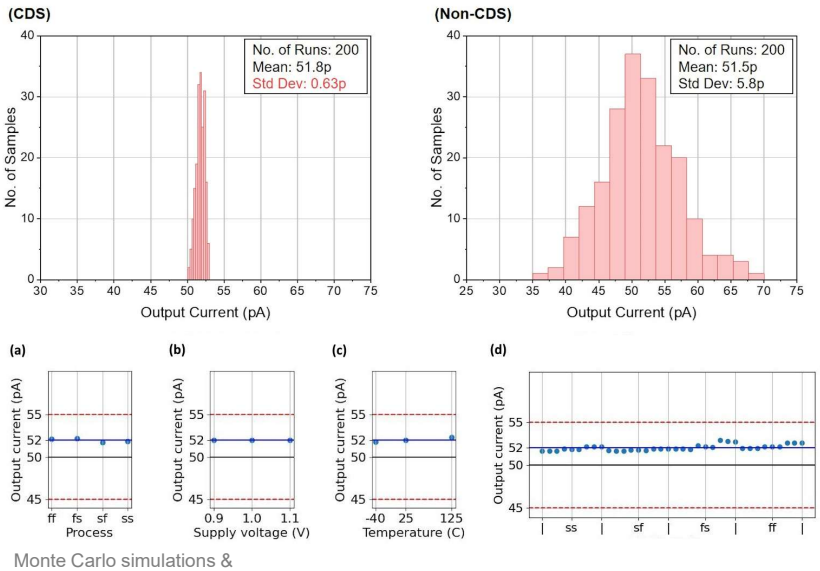
Directly control 'w', without individually setting V_g and V_{g_ref}

RESULTS



- Input current, 100fA – 2nA or 10nA
- Measuring output current of a single cell
- (a) Single cell I_{out} (b) Error to ideal multiplication (c) Percentage error (d) Logarithmic Percentage Error

PVT VARIATION



Monte Carlo simulations &
Process (slow-slow, fast-slow, slow-fast, fast-fast), voltage and temperature corners

CONCLUSION

- Increasing PVT robustness allows for larger scalability with smaller compromise in performance
- The proposed CDS-based programming scheme essentially self calibrates the circuit
- Periodically reprogramming the weights with the CDS scheme, can account for temperature variations in real time.

The scalability of the design is further enhanced by:

- Small cell design: 2 transistors and one small capacitor (5fF), in a 22nm CMOS process
- High power efficiency (0.5fJ per MAC)
- Row parallel weight programming

REFERENCES

[1] S. Chakrabartty and G. Cauwenberghs, "Sub-microwatt analog VLSI trainable pattern classifier," IEEE Journal of Solid-State Circuits, vol. 42, no. 5, pp. 1169–1179, 2007.

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